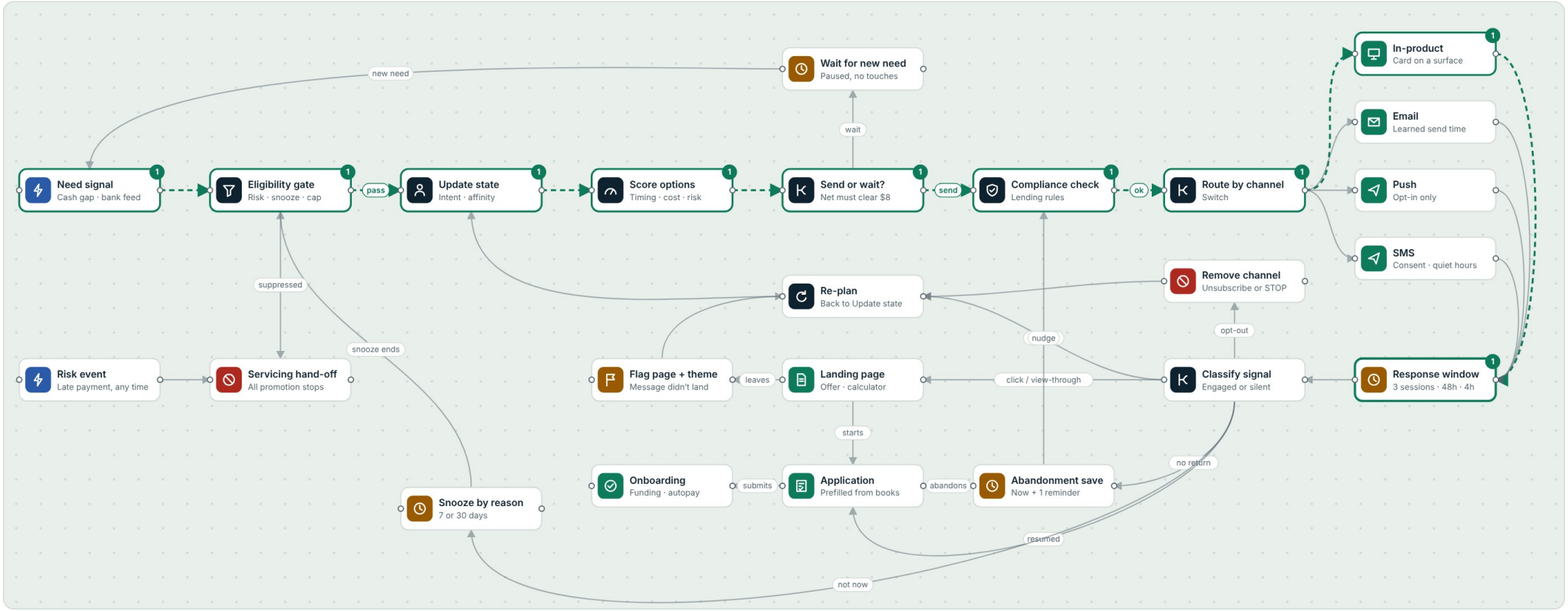
The journey as a workflow

An adaptive next-best-action engine for small-business lines of credit, term loans and credit cards, shown node by node

Rakesh R. Iyer

Concept demo · synthetic data · not affiliated with Intuit

One loop, from need signal to response



Trigger

Greenline's forecast shows an $18K gap before Nov 20

Engine

Gate, update state, score, decide, check compliance, route

Result

In-product card at 7 AM, this owner's strongest hour

How to read the workflow

Trigger
Need signals and risk events that start or interrupt the journey

Logic
Gate, state update, scoring, send-or-wait, routing, classification

Action or outcome
Channels, landing page, application, onboarding

Wait or learn
Response windows, abandonment save, snooze, page flags

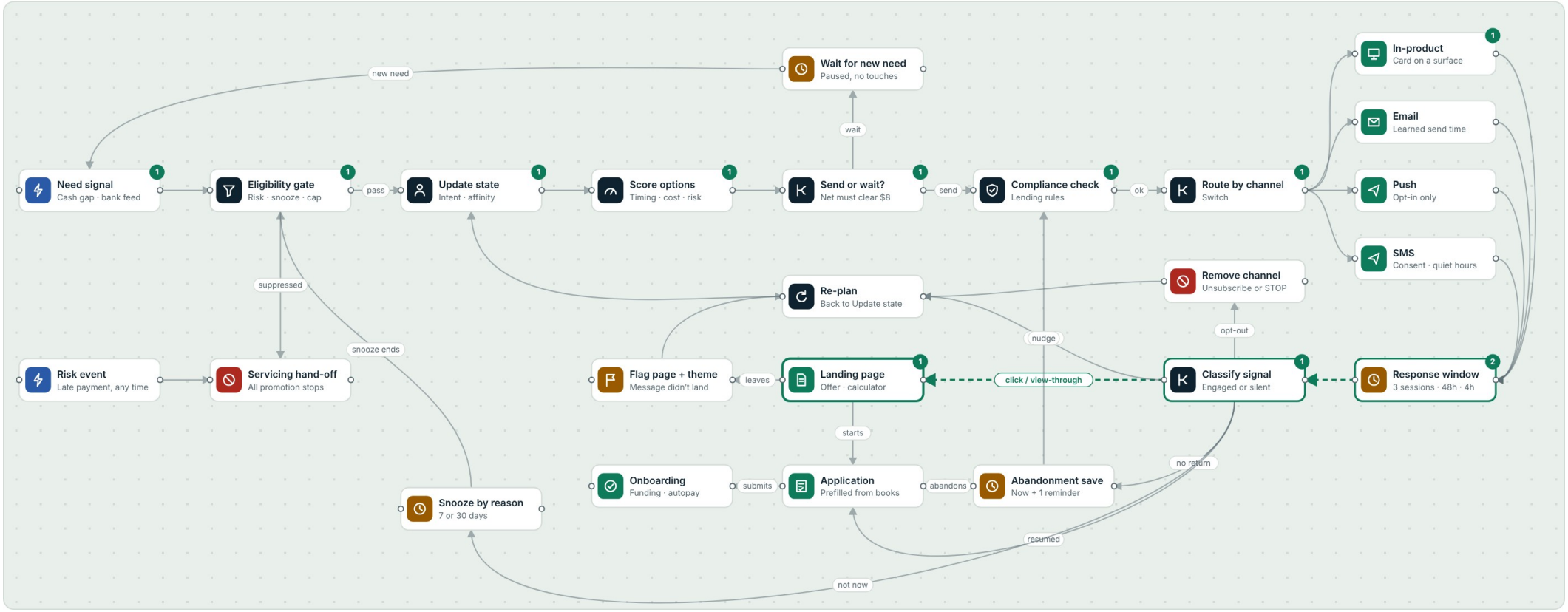
Stop
Opt-outs and the Servicing hand-off end promotion

The adaptive loop

1. Sense: capture clicks, visits and silence
2. Interpret: update intent, affinity, concern
3. Decide: channel, time, message, or wait
4. Act: send within caps and consent
5. Observe: wait out the response window

Dashed green = the path that just ran · pulsing node = where the customer is now · badge = times a node ran

A click lands on the landing page



Signal

Clicked the in-product CTA

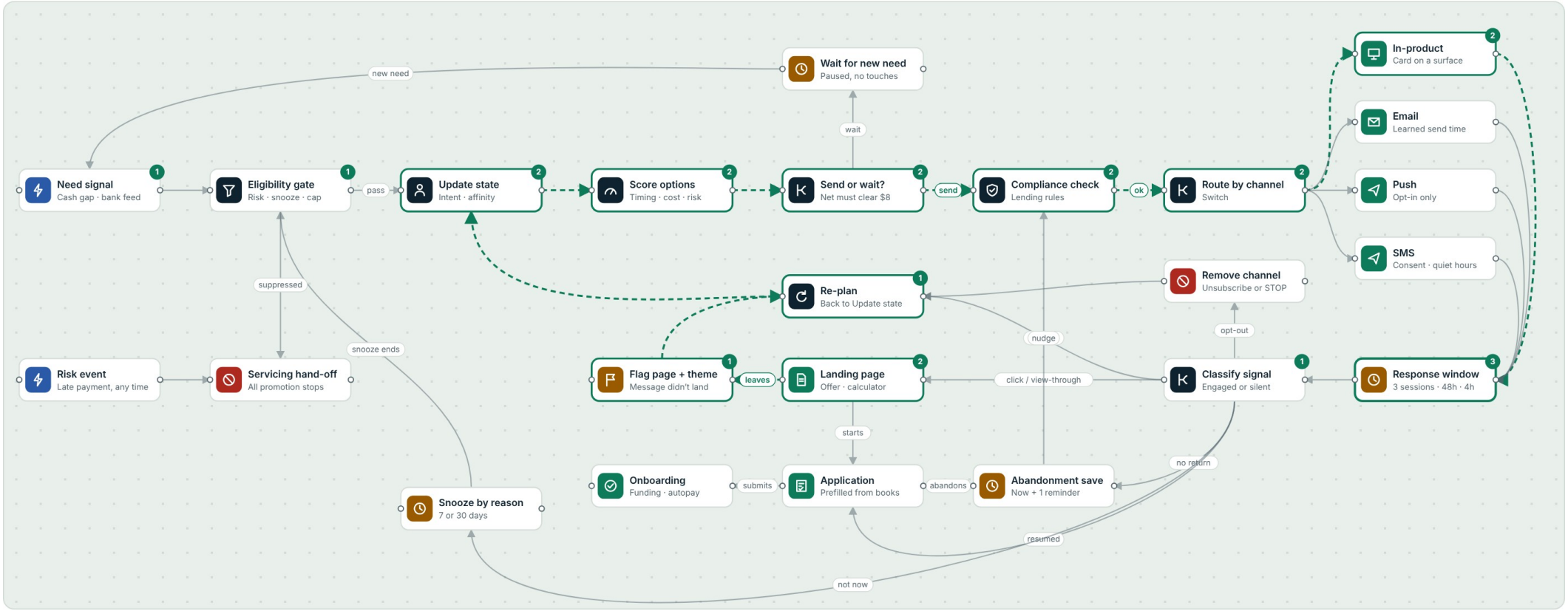
Engine

Intent +20, in-product affinity +0.15

Next

Hold all outbound while the customer is in flow

Leaving the page re-plans the message



Signal

Left the landing page without taking action

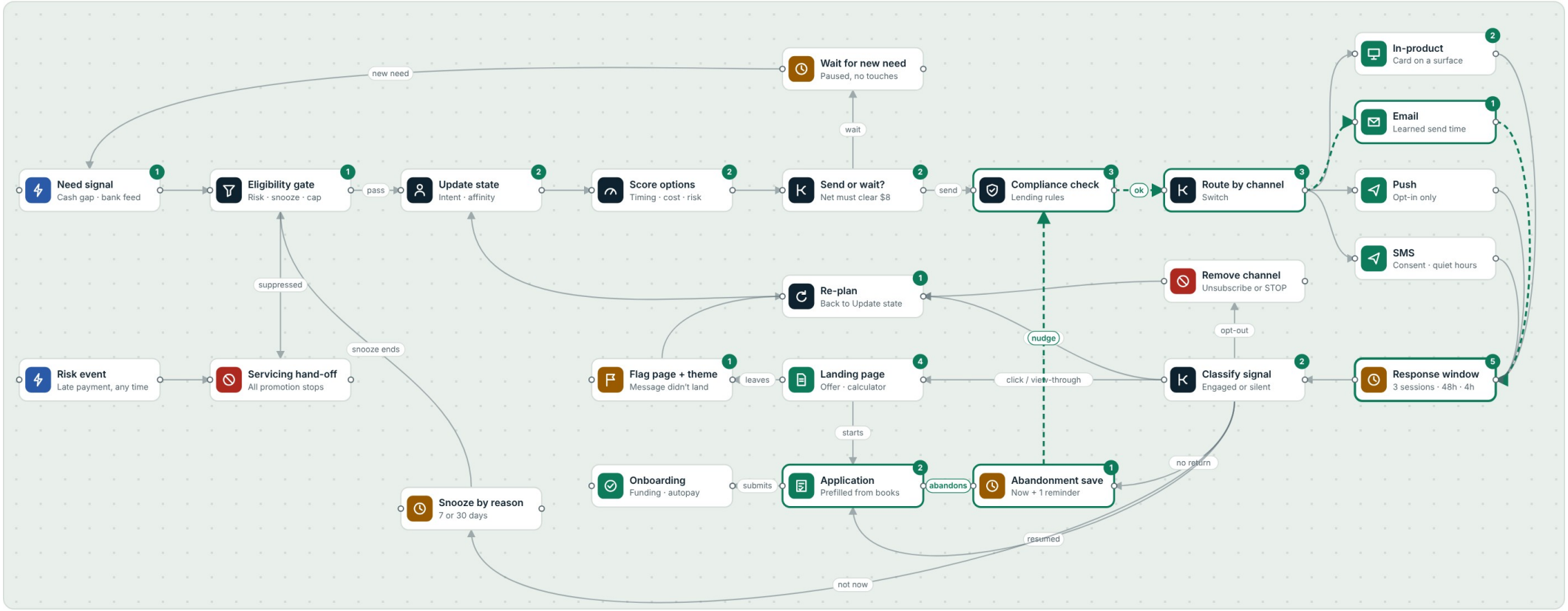
Engine

Theme marked as not landing; page flagged for a test

Next

Cost message with a calculator, next day at 7 AM, not instantly

An abandoned application gets an immediate save



Signal

Started, then abandoned the application

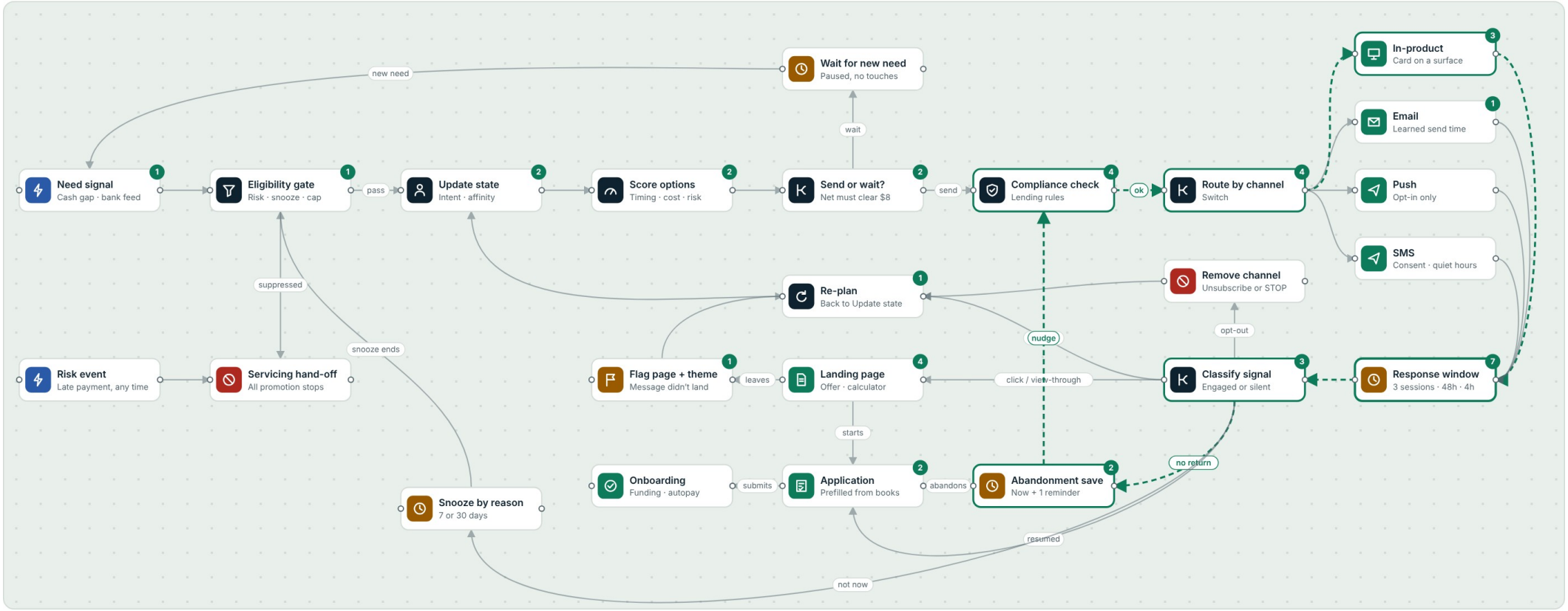
Engine

Abandonment branch: service moment, no cadence cost

Next

Email within 15 minutes: “Your application is saved”

One reminder inside 24 hours, on a new channel



Signal

No return after the email nudge

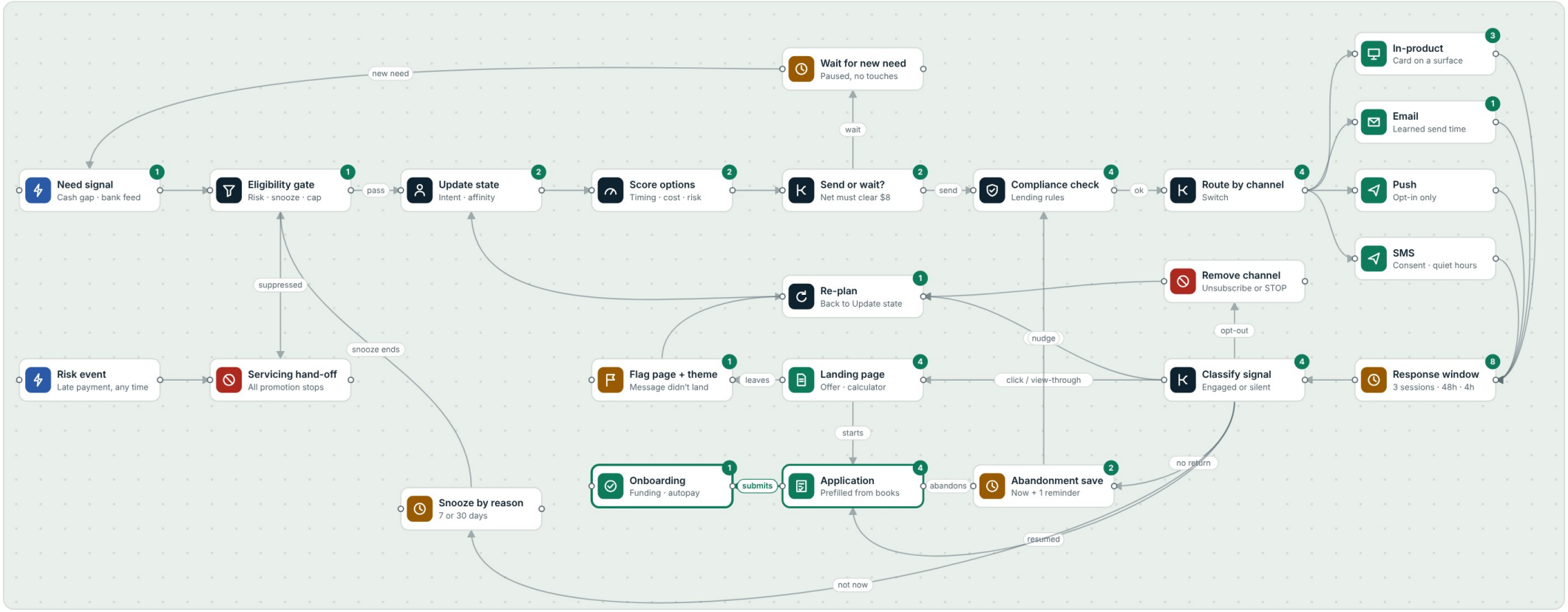
Engine

Picks a different channel with a slot before the deadline

Next

In-product “Resume application” at 6 PM the same day

Submitted: hand off to onboarding



Signal

Came back and submitted the application

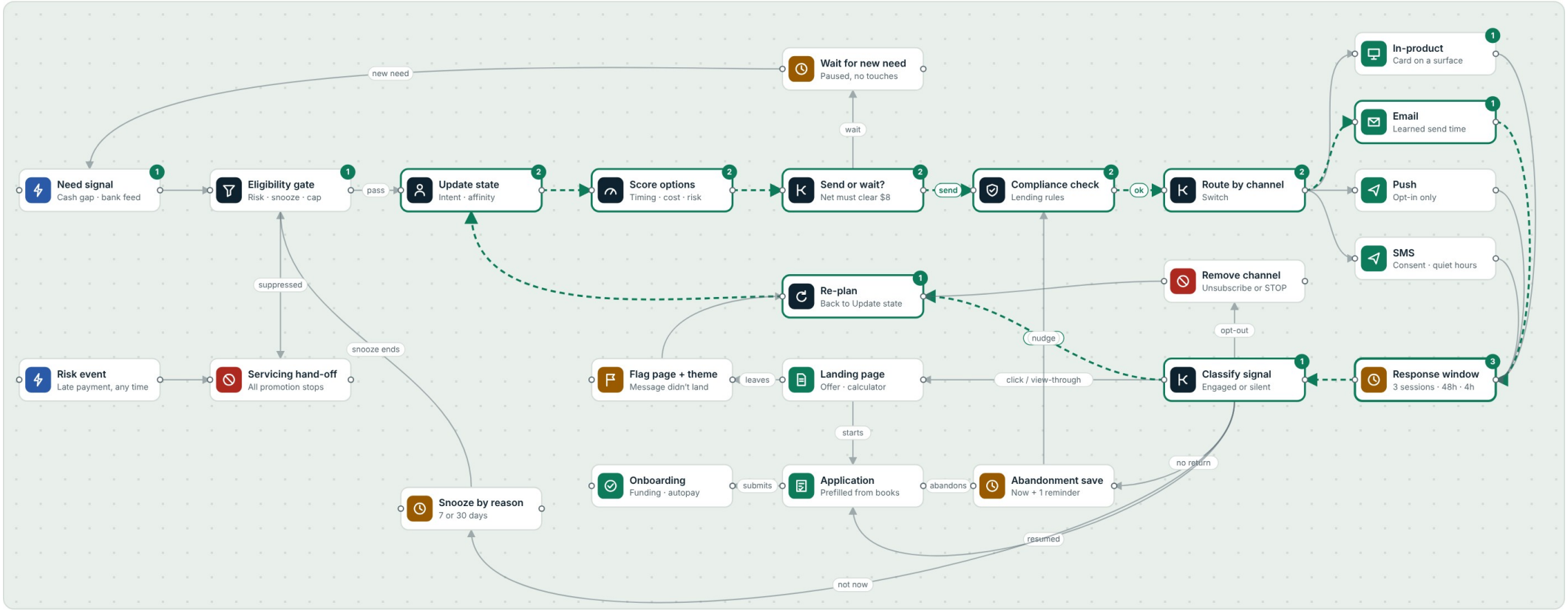
Engine

This journey ends

Next

Onboarding journey: funding, autopay, first draw

Silence switches the channel



Signal

Ignored the in-product card for 3 sessions

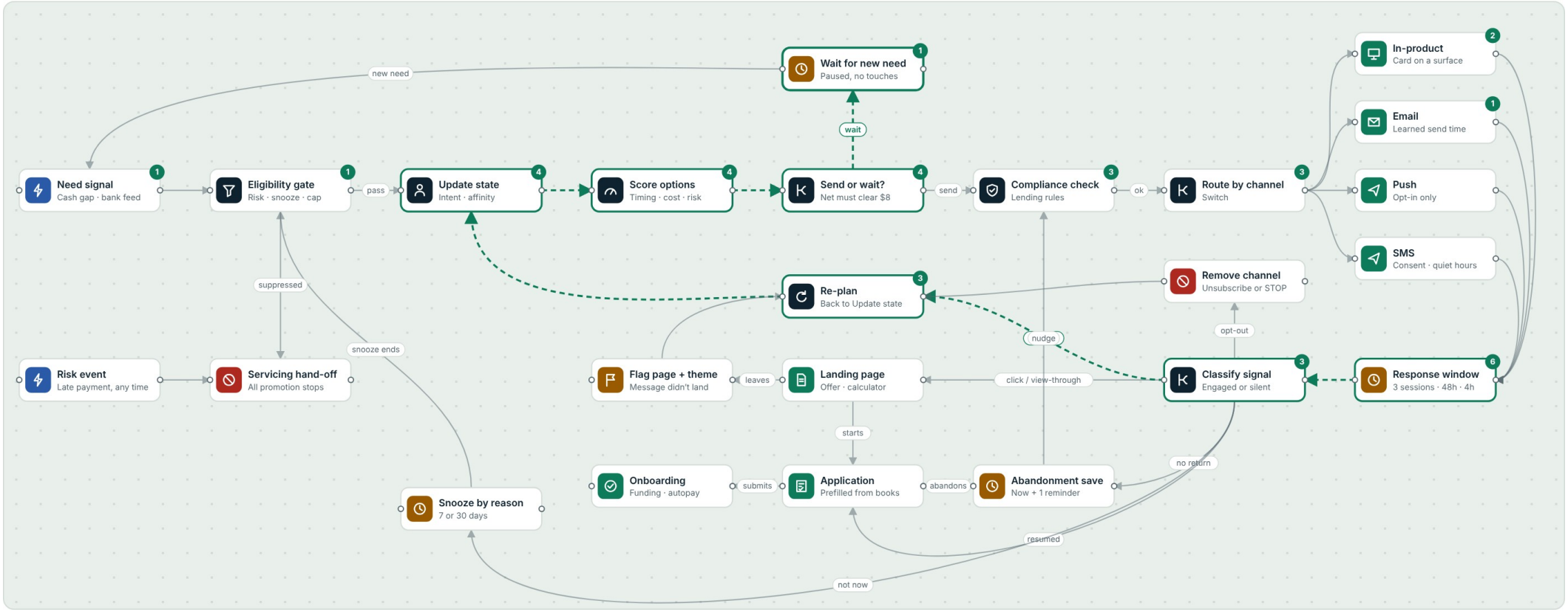
Engine

In-product affinity −0.15; theme rotates

Next

Email with a cost message at a strong evening slot

Three misses in a row: the engine waits



Signal

No click on email, then another ignored card

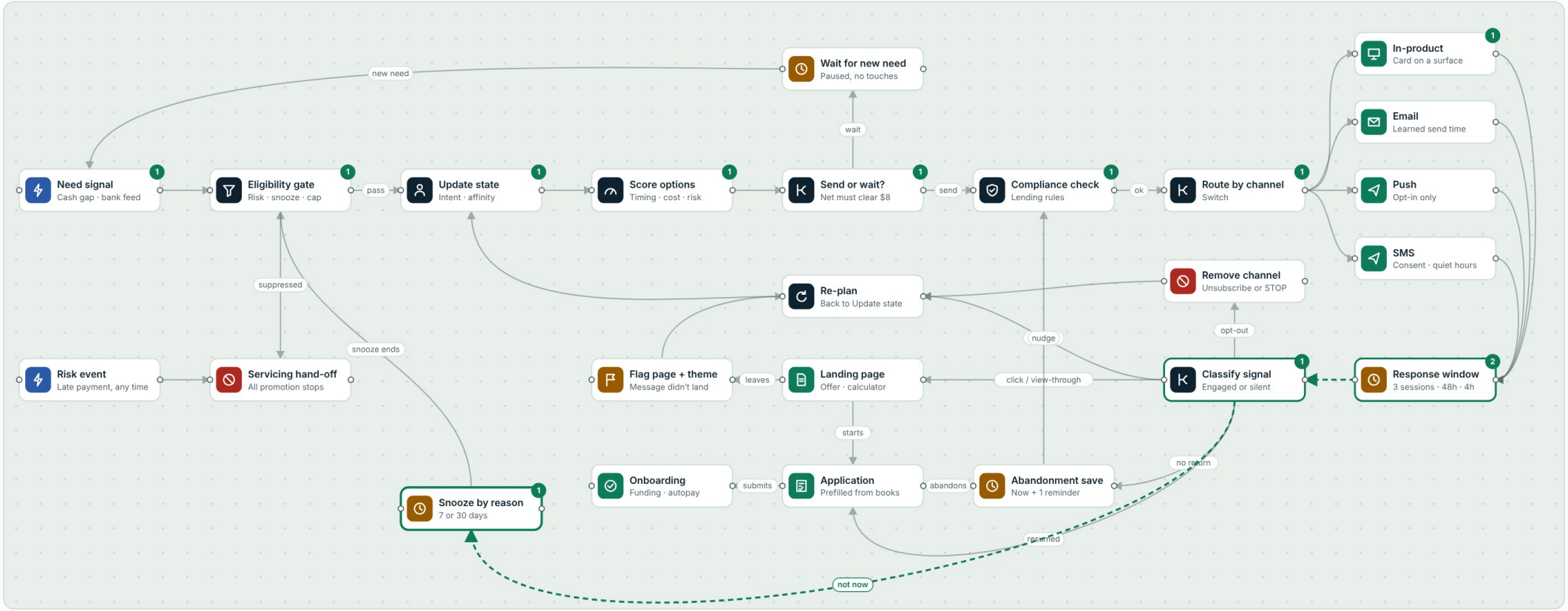
Engine

Three unanswered touches; opt-out risk rising

Next

Wait for a new need signal instead of adding noise

“Not now” snoozes by reason



Signal

“Not now: worried about the cost”

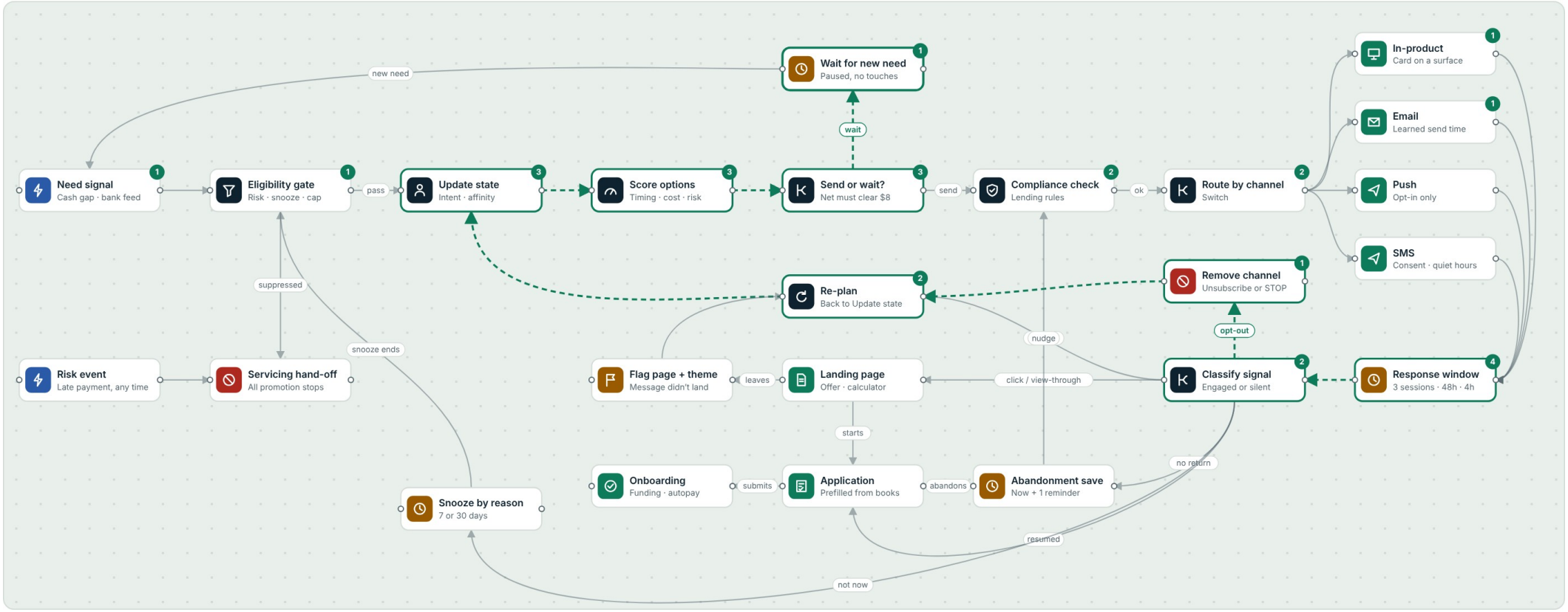
Engine

7-day snooze; next message will lead with cost

Next

Snooze ends, or a new need ends it early

An opt-out removes the channel for good



Signal

Unsubscribed from email

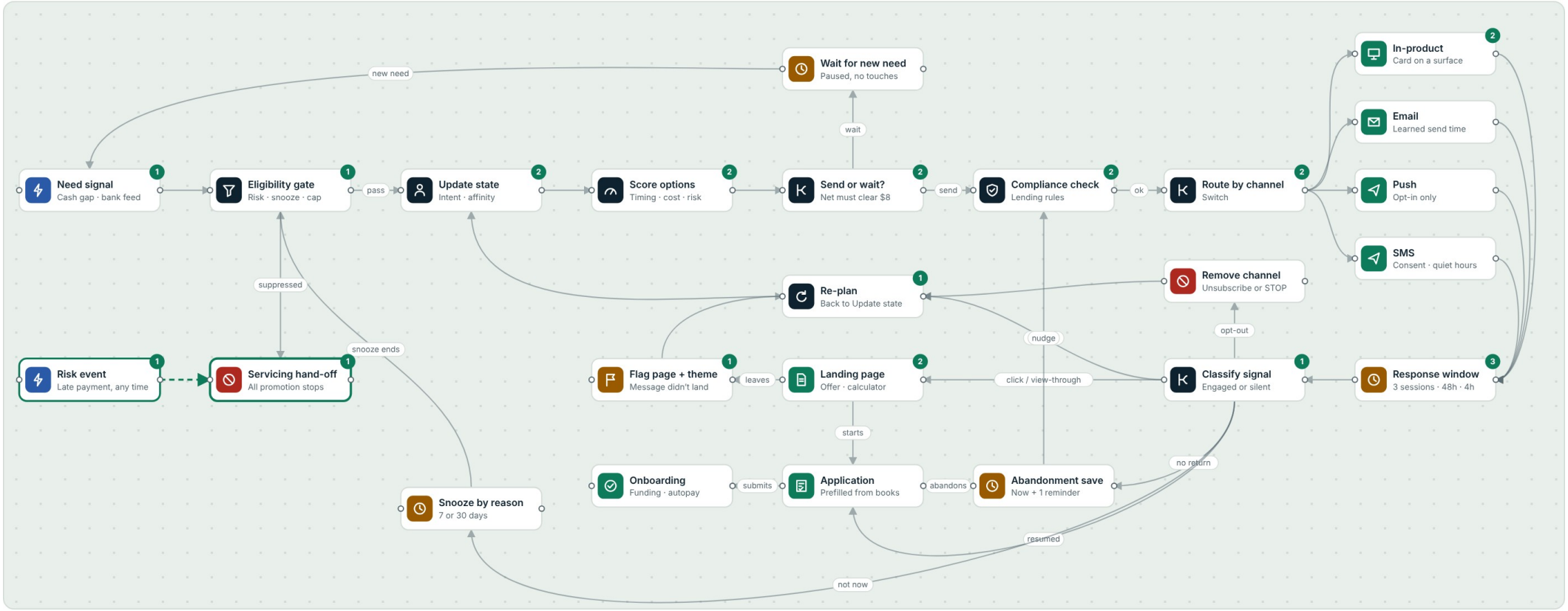
Engine

Email removed; about $810 of 12-month channel value lost

Next

Re-plan on remaining channels, or wait

A risk event overrides every stage



Signal

Late payment reported on an existing account

Engine

Suppress all Capital and card promotion

Next

Servicing takes over with one shared suppression list

Every action or inaction re-enters the loop

Explainable decisions on synthetic data today; with real behavioral signals, Data Science models and a holdout, the same workflow scales

Rakesh R. Iyer

Concept demo · synthetic data